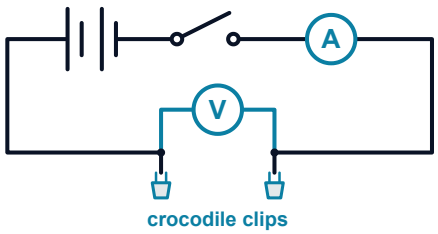
DO NOW · REQUIRED PRACTICAL

Resistance and the length of a wire

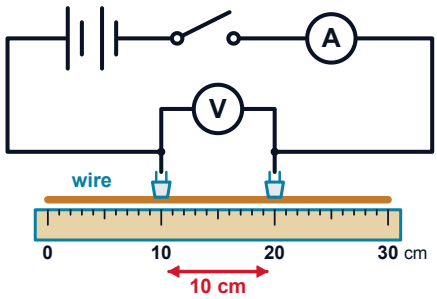
Electricity
Paper 1 · Combined and Separate
6 steps

A student investigates how the length of a wire affects its resistance.

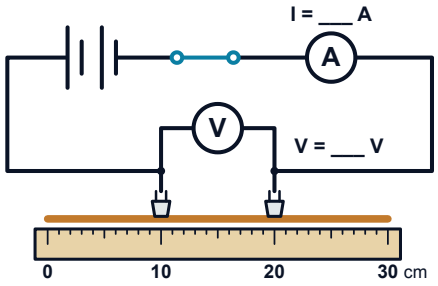
1 Set up a circuit with an ammeter in series, ending in two crocodile clips. Connect a voltmeter across the clips.



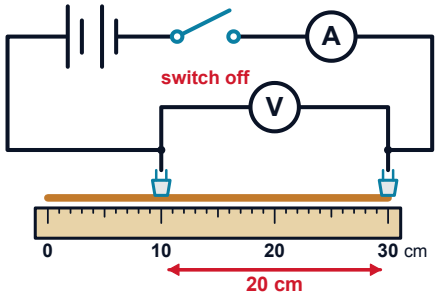
2 Attach the crocodile clips 10 cm apart on the wire, so the voltmeter is across this length of wire.



3 Switch on and record the current and the potential difference.



4 Switch off to let the wire cool, then move the clips to 20 cm apart.



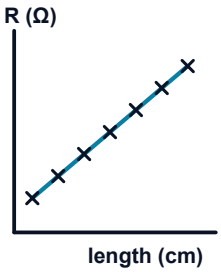
5 Repeat the readings for a range of lengths.

length (cm)	I (A)	V (V)
5		
10		
15		
20		
25		
30		

6 Calculate the resistance for each length and plot resistance against length.

$$R = V \div I$$

(for each length)



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